

Motion-Aware Watch-Only Stress Recognition with Privileged Physiological Supervision

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Stress recognition from watches could enable low burden affective monitoring, but many wearable studies are conducted with richer physiological sensors than are available during everyday deployment. This creates an asymmetric sensing problem: training may observe research-grade or chest-worn physiology, whereas inference is restricted to watch PPG/BVP and wrist accelerometry. This motivates examining how wrist ACC should interact with motion-sensitive PPG/BVP signals, and whether privileged physiological supervision consistently improves a constrained watch-only model. We introduce a Motion-aware watch encoder that retains pulse as the primary physiological pathway and uses ACC only as structured context through bounded feature Adaptation and motion-conditioned Correction, and use this same deployable encoder to evaluate whether training-time privileged physiological supervision improves watch-only inference. We evaluate the framework on Galaxy PPG and WESAD using subject-disjoint leave-one-subject-out protocols. The experiments first compare alternative ways of incorporating wrist ACC into a pulse-based deployable model under the same downstream backbone, and then compare different privileged-supervision objectives on the same Motion-aware watch-only encoder. Across both datasets, unrestricted pulse+ACC concatenation degrades validation-thresholded performance relative to Pulse-only sensing, whereas Motion-aware conditioning yields higher mean balanced accuracy and F1 while keeping pulse as the primary physiological pathway. Privileged physiological supervision is also not uniformly beneficial. On Galaxy PPG, Standard KD increases AUROC by 0.027 but reduces balanced accuracy and F1 by 0.050 and 0.086, respectively; confidence-weighted distillation partially mitigates this degradation but does not surpass the non-distilled student on thresholded metrics. On WESAD, where the privileged teacher is stronger and more stable, uniform distillation largely preserves thresholded performance, while confidence gating is overly conservative. These results show that ACC is most useful as structured context for pulse interpretation, and that privileged physiology should be treated as conditional training guidance, not as an unquestioned oracle.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Computing methodologies** → *Neural networks*.

Additional Key Words and Phrases: wearable stress recognition, smartwatch sensing, privileged supervision, knowledge distillation

1 Introduction

Wearable stress recognition is easier to translate into everyday use when it relies on sensors that people already wear. Recent wearable-stress studies show that physiological stress recognition commonly uses multimodal biosignals, including cardiovascular, electrodermal, respiratory, thermal, muscular, and inertial measurements [3, 16]. However, this multimodal research setting does not always match routine smartwatch deployment: training may observe rich physiological signals collected in controlled studies, whereas the deployed model may be limited to watch-available pulse and motion signals. This creates an asymmetric sensing problem between rich study-time observation and constrained everyday inference.

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This asymmetry first raises the question of how the retained watch signals should be represented. PPG/BVP provides pulse-related cardiovascular evidence, but wrist pulse measurements are sensitive to motion, contact, and sensing conditions [5]. Prior PPG-ACC work shows that accelerometry can help interpret motion-sensitive pulse signals, especially in heart-rate estimation settings [11, 20]. In stress recognition, however, wrist ACC is not straightforwardly another physiological stress channel. It may describe movement, activity, posture, or signal quality, all of which can interact with stress-related pulse variation. This makes ACC integration a modeling choice, not a simple question of whether another sensor is available: ACC may be used as an unrestricted predictive input, or it may be used as context for interpreting pulse evidence.

The same training-deployment asymmetry also raises a second question: how should richer physiological signals be used when they are available only during training? Privileged information and distillation methods provide a natural framework for transferring training-time information to a student model with a restricted inference input space [8, 13]. Yet in wearable physiology, a richer teacher is not automatically an oracle for a watch-only student. Teacher and student signals may differ in body location, sensor technology, physiological pathway, and signal quality, so a teacher can rely on evidence that the deployed watch representation cannot reproduce [14]. Recent cross-modal distillation work suggests that transfer depends on the relationship between teacher and student modalities rather than on teacher strength alone [26]. PULSE recently studied a closely related privileged-sensor transfer problem, distilling from a rich EDA teacher into deployable ECG, BVP, ACC, and temperature students without using EDA at inference [30]. We therefore study whether privileged physiological supervision improves the same constrained watch-only model that will be used at inference.

To address these questions, we introduce a Motion-aware watch encoder for deployment-constrained stress recognition. The encoder keeps PPG/BVP as the primary physiological pathway and uses wrist ACC as structured motion context rather than as an unrestricted parallel prediction branch. ACC-conditioned adaptation and feature-level correction allow motion information to modulate the pulse representation, while the final deployed model remains restricted to watch-available inputs. Using this same deployable encoder, we then evaluate whether training-time privileged physiological supervision improves watch-only inference, comparing non-distilled training with uniform and confidence-weighted distillation objectives.

We evaluate the framework on Galaxy PPG, a semi-naturalistic consumer-watch dataset, and WESAD, a controlled multimodal wearable benchmark, using subject-disjoint leave-one-subject-out protocols. The experiments are organized around two controlled comparisons. First, we compare alternative ways of incorporating wrist ACC into a pulse-based watch model under the same downstream backbone. Second, we hold the deployed Motion-aware architecture fixed and vary only the privileged-supervision objective. Across both datasets, unrestricted pulse-ACC concatenation underperforms the corresponding pulse-only baseline on validation-thresholded metrics, whereas Motion-aware conditioning provides stronger balanced accuracy and F1. Privileged physiological supervision is also conditional and not uniformly beneficial: teacher guidance can improve score ranking while weakening validation-thresholded behavior, depending on the dataset and distillation objective. This work makes these contributions:

- We introduce a Motion-aware watch representation that treats ACC as bounded context for pulse adaptation and correction rather than as an unrestricted stress-prediction branch.
- We evaluate privileged physiological supervision on the same deployable Motion-aware student, showing that teacher guidance can improve ranking while degrading validation-thresholded behavior and should therefore be treated as conditional training guidance, not as an oracle.

- We provide a subject-disjoint evaluation across two complementary settings: a semi-naturalistic consumer-watch dataset and a controlled multimodal wearable benchmark.

2 Related Work

2.1 Wearable Stress Recognition and Multimodal Sensing

Wearable stress recognition aims to infer stress-related affective and physiological states from unobtrusive body-worn sensors. Prior studies have used a broad range of biosignals, including ECG- or PPG-derived heart-rate and heart-rate-variability features, blood volume pulse, electrodermal activity, respiration, skin temperature, electromyography, and inertial measurements. Recent research shows that EDA and PPG are among the most frequently used wearable biosignals for stress detection, while multimodal settings are often preferred because no single physiological channel uniquely identifies psychological stress across users, activities, and sensing conditions [3, 4, 16, 23].

The move toward multimodal sensing is not only a matter of adding more sensors. It also reflects a central modeling difficulty: stress-relevant evidence is distributed across physiological systems, while activity, posture, and sensing conditions can alter the same signals. Multimodal benchmarks such as WESAD therefore synchronize wrist and chest measurements, including BVP, ECG, EDA, EMG, respiration, temperature, and acceleration, so that stress-related information can be studied across body locations and sensing modalities [21]. More recent datasets extend this direction by collecting wearable physiological signals under both stress and exercise conditions, or by combining physiological measurements with behavioral modalities such as facial expression [9, 10, 15]. These datasets highlight that multimodal stress recognition is not merely a matter of collecting more channels. Once physiological, behavioral, and motion signals are observed together, the modeling problem becomes how to represent their relationships under shared stress labels, especially when the same signal can reflect both stress-related physiology and non-stress factors such as movement or task context.

This issue becomes more pronounced under everyday deployment constraints. Chest ECG, respiration belts, EMG, and research-grade electrodermal sensors can be useful in data collection, but they are often unavailable or burdensome in routine smartwatch use. Consumer watches more commonly retain PPG/BVP and wrist accelerometry, and even these signals can reflect both stress-related physiology and non-stress factors such as movement, posture, sleep loss, or excitement. Prior work on context-aware stress detection shows that activity and context can improve wearable stress recognition, while consumer smartwatch stress scores may not map cleanly onto subjective psychological stress [22, 24]. This motivates separating deployable signals from training-only information: watch PPG/BVP and wrist ACC require a representation that specifies how motion should interact with pulse evidence, whereas richer physiological streams may provide privileged supervision during training but cannot define the deployed inference model.

2.2 Deployment-Retained Signals: PPG-ACC Fusion and Motion-Aware Pulse Modeling

When multiple deployable signals are available, wearable models often rely on generic multimodal fusion strategies that aggregate features or learn modality weights. Such strategies address how information from different channels can be combined, but they do not by themselves define the functional role assigned to each modality [1]. This distinction is important for watch-based stress recognition because PPG/BVP and wrist ACC are both available at inference, yet they do not measure the same physiological process: PPG/BVP provides pulse-related cardiovascular evidence, whereas ACC primarily describes wrist motion, activity, and sensing context.

The relationship between pulse and motion has been studied most extensively in PPG-based heart-rate estimation. Wrist PPG is attractive for low-burden cardiovascular monitoring, but it is sensitive to arm motion, contact pressure, sensor displacement, and optical coupling [2]. Classical PPG-ACC methods therefore use accelerometry as a motion-reference signal, as in TROIKA and joint sparse spectrum reconstruction, which combine wrist PPG with acceleration information to track heart rate during physical activity [28, 29]. The PPG-DaLiA dataset, introduced with Deep PPG, further supports learned PPG-ACC modeling under daily-life activities [20]. More recent work has moved toward learned pulse-motion representations, including convolutional models, cross-attentional feature fusion, and deployment-oriented modality-wise fusion for PPG-based heart-rate estimation [11, 12, 20]. Together, this line of work establishes that ACC is informative for interpreting wrist PPG under motion, but it is mainly optimized for recovering or tracking cardiac rhythm.

Stress recognition poses a different modeling problem. The goal is not only to recover a cleaner cardiac frequency or reconstruct a less contaminated pulse waveform, but to preserve stress-relevant physiological variation while accounting for movement, activity, and sensing conditions. Prior wrist-BVP stress-recognition work shows that pulse signals alone can support stress classification, but it does not resolve how accelerometry should enter a deployment-constrained pulse model [18]. In this setting, ACC can be modeled as an additional predictive modality, as a learned fusion partner, or as contextual information that conditions the interpretation of pulse evidence. Recent context-aware wearable fusion work similarly uses motion-related signals to characterize noise context in mobile health sensing [19]. Existing PPG-ACC fusion studies therefore motivate the use of accelerometry, but they do not fully resolve how wrist motion should enter a deployable pulse-based stress-recognition model. This motivates comparing ACC-integration strategies under the same downstream watch backbone, instead of treating PPG-ACC fusion as a single design choice.

2.3 Deployment-Removed Signals: Privileged Supervision and Distillation

Multimodal wearable studies often collect physiological streams that are absent during routine smartwatch inference, such as chest ECG, respiration, EDA, EMG, temperature, or research-grade BVP. Although these signals may provide useful stress-related evidence, treating them as ordinary fusion inputs would overstate the information available to the deployed model. In deployment-constrained stress recognition, they are better viewed as training-only information that can guide learning but must be removed from the final watch-only inference pathway.

This setting is closely related to learning with privileged information, where additional features are available during training but not at test time [25]. Knowledge distillation provides another teacher-student formulation, in which a student learns from a teacher’s softened predictive distribution rather than only from hard labels [8]. Generalized distillation connects these ideas by framing privileged information as a source of teacher supervision for a student with a restricted input space [13]. Cross-modal distillation extends this principle to settings in which teacher and student observe different modalities, for example by transferring supervision from a richer or better-labeled modality to a modality that remains available at deployment [6, 7]. Similar training-deployment separations have recently been studied in affect modeling, where laboratory-only modalities are used as privileged information while the deployed model relies on modalities available in the wild [14].

However, privileged physiological supervision is not equivalent to an oracle label source. In wearable stress recognition, the teacher and student may differ not only in model capacity, but also in body location, sensor technology, physiological pathway, and signal quality. A teacher using chest ECG, EDA, respiration, EMG, or research-grade BVP may classify stress well because it observes evidence that a watch PPG/BVP-ACC student cannot reproduce. More generally, recent analyses of cross-modal distillation show that knowledge transfer is not uniformly beneficial and

depends on the relationship between teacher and student modalities [26]. Confidence-aware distillation methods further suggest that teacher signals may need to be weighted when their reliability varies across examples [27]. In physiological sensing, however, teacher confidence reflects reliability in the privileged sensing space, not necessarily compatibility with the watch-only representation.

This motivates evaluating privileged physiology as conditional training guidance without assuming that richer sensing always provides an oracle target. In a deployment-constrained stress-recognition setting, the key question is whether teacher supervision improves the same watch-only student that will be used at inference. We therefore separate the deployable architecture from the training-time supervision source: privileged physiological streams may provide uniform or confidence-weighted distillation during training, but the privileged encoder, teacher classifier, auxiliary targets, and distillation losses are removed after training. This framing allows privileged supervision to be assessed by its effect on the fixed deployed watch model.

3 Method

We study watch-only stress recognition under asymmetric sensing constraints: training may use richer physiological measurements, whereas inference is restricted to pulse and accelerometry from the target watch. The method therefore has two components. First, the deployable representation determines how ACC conditions the interpretation of PPG/BVP. Second, the training objective determines whether and how a privileged physiological teacher supervises that fixed deployable representation. Section 3.1 formalizes the problem and presents the system overview. Section 3.2 describes the Motion-aware watch encoder, in which pulse is the primary physiological pathway and ACC affects it through bounded adaptation and correction. Section 3.3 compares the privileged-supervision objectives evaluated in this study. Section 3.4 then specifies the joint losses, dataset-specific auxiliary supervision, and the watch-only inference path.

3.1 Problem Setting and System Overview

Let $y_i \in \{0, 1\}$ denote the calm or stress label for window i . The deployable model receives only watch-available inputs,

$$\mathbf{x}_i^w = \left(\mathbf{x}_i^{\text{pulse}}, \mathbf{x}_i^{\text{acc}}, \boldsymbol{\phi}_i^w, q_i \right), \quad (1)$$

where $\mathbf{x}_i^{\text{pulse}}$ is the Galaxy PPG or WESAD wrist-BVP waveform, $\mathbf{x}_i^{\text{acc}}$ is tri-axial wrist accelerometry, $\boldsymbol{\phi}_i^w$ contains watch-derived wavelet features, and q_i is the watch-quality variable. For Galaxy PPG, q_i is derived from the device-reported PPG status flags; for WESAD, $q_i = 1$ for every window.

The Motion-aware watch encoder f_w maps the deployable input to a representation $\mathbf{h}_i^w$, which is passed to the watch classifier C_w :

$$\mathbf{h}_i^w = f_w(\mathbf{x}_i^w), \quad (2)$$

$$\mathbf{p}_i^w = \text{softmax}(C_w(\mathbf{h}_i^w)). \quad (3)$$

This watch pathway is the complete inference model. It is retained unchanged across the privileged-supervision objectives and never requires privileged sensors at deployment. During training, some examples additionally provide privileged physiological signals $\mathbf{x}_i^p$. A teacher encoder f_T and classifier C_T produce

$$\mathbf{h}_i^T = f_T(\mathbf{x}_i^p), \quad (4)$$

$$\mathbf{p}_i^T = \text{softmax}(C_T(\mathbf{h}_i^T)). \quad (5)$$

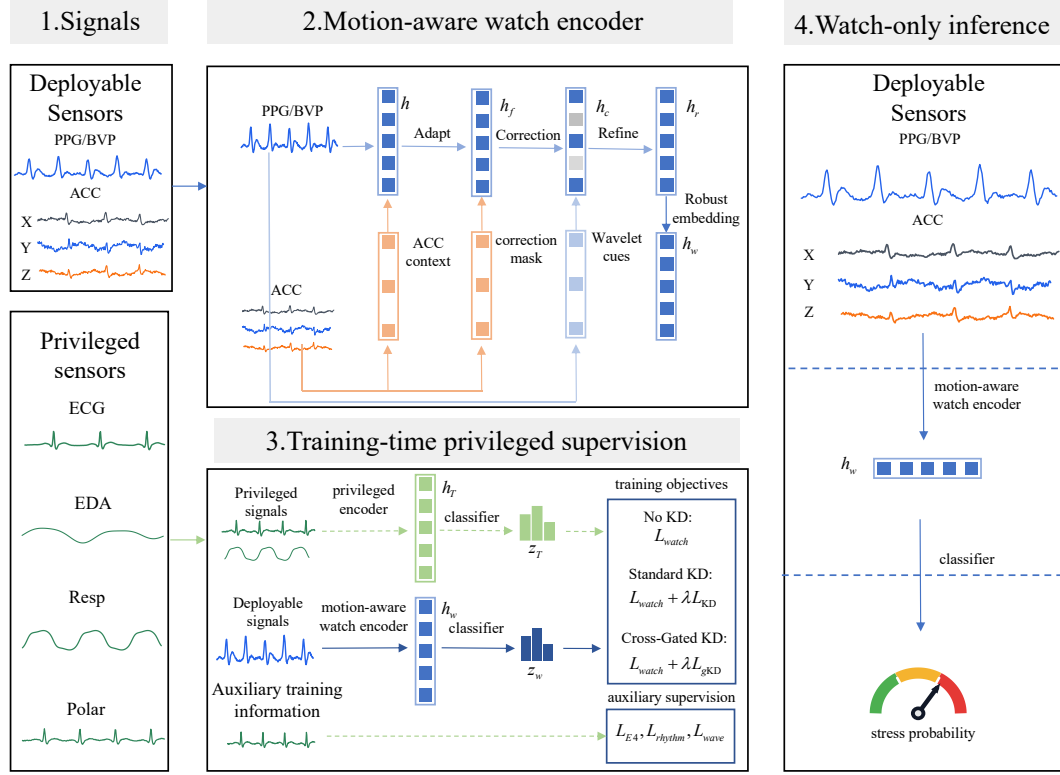


Fig. 1. Overview of the watch-only stress-recognition framework. The deployable pathway receives watch PPG/BVP, ACC, wavelet features, and the watch-quality variable q . Its Motion-aware encoder treats pulse as the primary physiological pathway and uses ACC through bounded adaptation and motion-conditioned correction. During training, a privileged encoder may produce teacher logits from richer physiological sensing. The non-distilled condition, Standard KD, and Cross-Gated KD are evaluated as parallel training objectives; they are not selected by an automatic regime-diagnosis mechanism. At inference, the privileged encoder, teacher classifier, auxiliary targets, and distillation losses are removed, leaving the same watch-only pathway for all compared objectives.

Figure 1 summarizes the training and inference pipeline. The deployable watch pathway is the only path used at inference: it receives watch pulse, wrist ACC, watch-derived wavelet features, and the watch-quality variable, and outputs the stress probability through the Motion-aware encoder and watch classifier. The Motion-aware encoder uses ACC-conditioned Adaptation and Correction to modulate the pulse representation, then applies wavelet/quality-guided Refinement and temporal embedding before classification. During training, an optional privileged teacher may process richer physiological signals and provide distributional supervision to the same watch pathway. Dataset-specific auxiliary targets may also supervise intermediate training objectives when available. These privileged sensors, teacher computations, auxiliary targets, and distillation losses are removed after training.

The three privileged-supervision objectives are treated as parallel training alternatives. The non-distilled condition trains the watch pathway without teacher-to-student distribution transfer. Standard KD applies teacher supervision

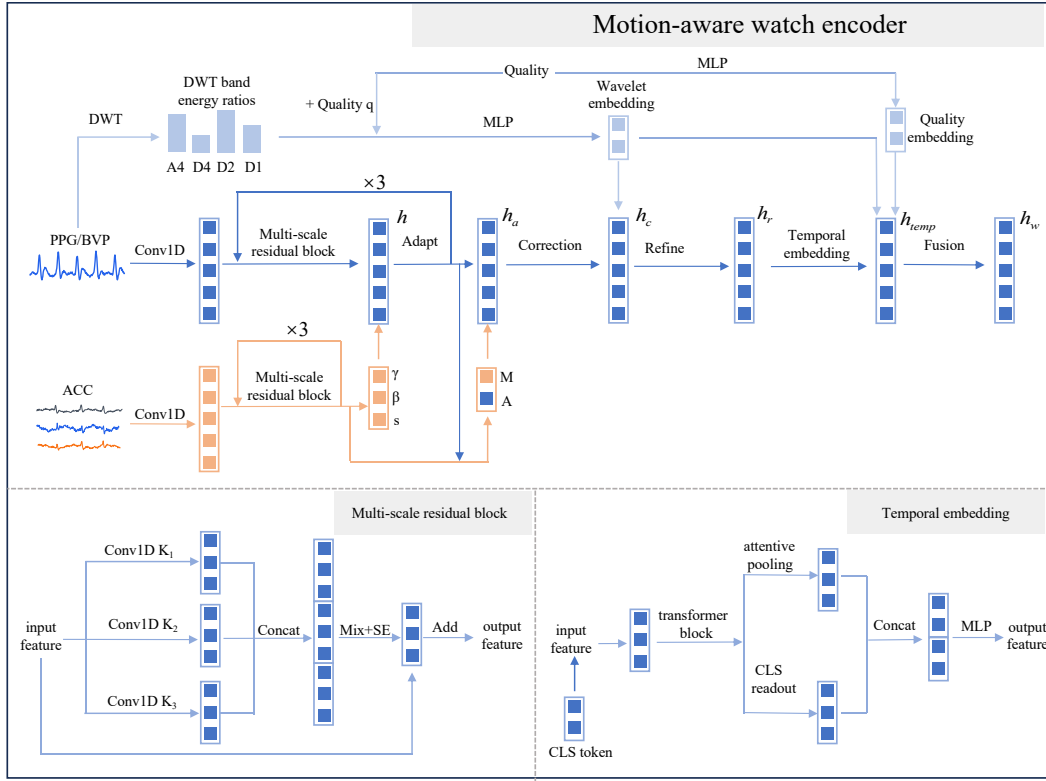


Fig. 2. Motion-aware watch encoder. Separate PPG/BVP and ACC streams pass through multi-scale residual stages. ACC derived context performs bounded stage-wise adaptation of the pulse representation. A Correction module jointly analyzes the adapted pulse and projected ACC features to produce a correction mask and a signed residual basis. Wavelet-derived pulse cues and the watch quality variable q subsequently guide Refinement. A temporal embedding module combines Transformer CLS readout and attentive pooling, and the resulting representation is fused with the wavelet and quality embeddings to produce the deployable watch representation h^w .

uniformly across training windows. Cross-Gated KD weights teacher supervision using watch quality together with teacher and stop-gradient student confidence.

After training, the model predicts each watch window using only the watch pathway:

$$\hat{y}_i = \mathbb{I} [p_i^w(y = 1) \geq \delta], \quad (6)$$

where $p_i^w(y = 1)$ is the watch-only probability of the stress class, and δ is selected on subject-disjoint validation participants and then applied unchanged to the held-out participant. All privileged inputs, teacher predictions, auxiliary targets, and teacher-side computations are removed at inference.

3.2 Motion-Aware Watch Encoder

The Motion-aware watch encoder is the deployable backbone used throughout the study. Its design follows a functional separation between the two watch modalities: PPG/BVP provides the primary cardiovascular evidence for stress recognition, whereas ACC characterizes the motion and sensing context under which that pulse evidence is observed. ACC is therefore not used as an independent stress-prediction pathway and is not directly concatenated with the final classification representation. Instead, ACC influences the pulse pathway through two bounded operations: stage-wise feature adaptation and motion-conditioned correction. The corrected pulse-centered representation is then refined using watch-derived wavelet cues and the watch-quality variable q , as illustrated in Figure 2. We denote the output of one multi-scale residual block applied to an input feature sequence $\mathbf{u}$ as $\text{MSR}(\mathbf{u})$. For branch j , the block first computes

$$\mathbf{b}_j = \text{SepConv}_{k_j, d_j}(\mathbf{u}), \quad (7)$$

where k_j and d_j denote the kernel size and dilation. The complete block output is

$$\text{MSR}(\mathbf{u}) = \text{GELU}(\text{SE}[\text{Mix}([\mathbf{b}_1; \mathbf{b}_2; \mathbf{b}_3])] + \mathcal{R}(\mathbf{u})). \quad (8)$$

The PPG/BVP blocks use kernel sizes (7, 15, 31), while the ACC blocks use (5, 11, 19). Both streams use dilation factors (1, 1, 2) with padding that preserves temporal length. The broader pulse receptive fields capture local pulse morphology and longer rhythm-related variation, whereas the ACC stream emphasizes comparatively local motion patterns.

Across the three residual stages, the ACC stream produces motion-context summaries for the pulse stream, but the pulse feature is not repeatedly overwritten by an adapted feature at every stage. Let $\mathbf{h}_\ell$ and $\mathbf{a}_\ell$ denote the pulse and ACC features at stage ℓ , respectively. The ACC feature is temporally pooled to obtain a stage-specific context vector,

$$\mathbf{c}_\ell = \text{Pool}(\mathbf{a}_\ell). \quad (9)$$

A FiLM-style module maps each $\mathbf{c}_\ell$ to channel-wise multiplicative and additive terms, $\boldsymbol{\gamma}_\ell(\mathbf{c}_\ell)$ and $\boldsymbol{\beta}_\ell(\mathbf{c}_\ell)$, with a learnable scalar $s_\ell \in (0, 1)$ bounding the strength of the corresponding ACC-derived modulation.

Pulse adaptation is applied after each of the three residual stages, using the stage-specific ACC context parameters.

$$\mathbf{h}^a = \mathbf{h} \odot [1 + s \tanh(\boldsymbol{\gamma}(\mathbf{c}))] + s\boldsymbol{\beta}(\mathbf{c}), \quad (10)$$

where $\mathbf{h}$ is the pulse feature before adaptation and $\mathbf{c}$ is the corresponding ACC context. This formulation lets the model apply bounded motion-conditioned modulation progressively at multiple depths before the final adapted pulse representation is passed to the subsequent correction module.

The adapted pulse feature is then passed to one motion-conditioned correction module at the highest feature level. Let $\mathbf{h}^a \in \mathbb{R}^{C_p \times T_p}$ denote the final adapted pulse feature and $\mathbf{a} \in \mathbb{R}^{C_a \times T_a}$ the final ACC feature. When their temporal lengths differ, the ACC feature is linearly interpolated to length T_p and projected into the pulse feature space:

$$\mathbf{a}^p = \mathcal{P}_{\text{acc}}(\mathbf{a}), \quad \mathbf{a}^p \in \mathbb{R}^{C_p \times T_p}, \quad (11)$$

where $\mathcal{P}_{\text{acc}}$ consists of a 1×1 Conv1D, batch normalization, and GELU activation. The adapted pulse feature, projected ACC feature, and their absolute discrepancy are concatenated:

$$\mathbf{J} = [\mathbf{h}^a; \mathbf{a}^p; |\mathbf{h}^a - \mathbf{a}^p|]. \quad (12)$$

Two lightweight heads process $\mathbf{J}$. A mask head produces a bounded correction mask $\mathbf{M} \in [0, 1]^{C_p \times T_p}$, and a residual head produces a signed correction basis $\mathbf{A} \in \mathbb{R}^{C_p \times T_p}$. The corrected pulse representation is

$$\mathbf{h}^c = \mathbf{h}^a - \tanh(\lambda_a) \mathbf{M} \odot \mathbf{A}, \quad (13)$$

where λ_a is a learnable scalar that bounds the overall correction strength. The mask determines where and to what extent the signed residual basis is applied; the module subtracts the bounded masked residual rather than the mask itself.

Neither $\mathbf{M}$ nor $\mathbf{A}$ is supervised using artifact annotations or clean-waveform targets. We therefore interpret Equation 13 as motion-conditioned feature correction rather than explicit physiological signal reconstruction or artifact removal.

The corrected feature is then refined using pulse-derived time–frequency cues. The watch-derived wavelet vector ϕ^w contains level-4 db4 band-energy ratios from the A_4 , D_4 , D_2 , and D_1 bands. This vector is combined with the watch-quality variable q and projected into a wavelet embedding $\mathbf{e}_\phi$; q is also separately projected into a quality embedding $\mathbf{e}_q$. A multi-scale residual block proposes a refinement candidate,

$$\tilde{\mathbf{h}}_r = \mathcal{R}_\phi(\mathbf{h}_c) \quad (14)$$

where $\mathcal{R}_\phi$ uses kernel sizes (5, 11, 21) and dilation factors (1, 2, 4). The wavelet embedding predicts bounded affine modulation terms, and a learnable residual update produces the refined pulse feature:

$$\bar{\mathbf{h}}_r = \tilde{\mathbf{h}}_r \odot \left(1 + 0.25 \tanh \gamma_\phi(\mathbf{e}_\phi) \right) + 0.25 \tanh \beta_\phi(\mathbf{e}_\phi), \quad (15)$$

$$\mathbf{h}_r = \mathbf{h}_c + \tanh(\lambda_\phi) (\bar{\mathbf{h}}_r - \mathbf{h}_c). \quad (16)$$

where λ_ϕ bounds the refinement strength. This formulation allows pulse-derived wavelet cues to adjust the representation without replacing the corrected pulse pathway.

Finally, the refined sequence $\mathbf{h}^r$ is projected into temporal tokens and augmented with a learnable CLS token and sinusoidal positional encoding. A Transformer encoder produces contextualized token features $\{\mathbf{u}_t\}$. The sequence is summarized using both CLS readout and attentive pooling:

$$\alpha_t = \text{softmax}_t(s(\mathbf{u}_t)), \quad (17)$$

$$\mathbf{h}^{\text{temp}} = \mathcal{H}_{\text{temp}} \left(\left[\mathbf{u}_{\text{CLS}}; \sum_t \alpha_t \mathbf{u}_t \right] \right). \quad (18)$$

The final deployable representation fuses the temporal, wavelet, and quality embeddings:

$$\mathbf{h}^w = \mathcal{H}_{\text{fuse}}([\mathbf{h}^{\text{temp}}; \mathbf{e}_\phi; \mathbf{e}_q]), \quad \mathbf{z}^w = C_w(\mathbf{h}^w). \quad (19)$$

The final fusion does not receive a separately pooled ACC embedding. Accelerometry affects prediction only through bounded Adaptation and Correction, so the inference path remains pulse-centered while retaining motion context throughout representation learning.

3.3 Privileged-Supervision Objectives

We evaluate privileged supervision by comparing the non-distilled condition with two teacher-guided conditions on the same Motion-aware watch encoder. The non-distilled condition trains the watch pathway without teacher-to-student distribution transfer. The teacher-guided conditions use either uniform distillation or confidence-weighted distillation.

Let z_i^T and z_i^w denote the teacher and watch logits for training window i . At temperature τ , the teacher and watch-student distributions are

$$p_i^{T,\tau} = \text{softmax}(z_i^T / \tau), \quad p_i^{w,\tau} = \text{softmax}(z_i^w / \tau). \quad (20)$$

The per-window distillation loss is

$$\ell_{\text{KD},i} = \tau^2 D_{\text{KL}}(p_i^{T,\tau} \parallel p_i^{w,\tau}). \quad (21)$$

We use $\tau = 4$. The non-distilled condition sets the distillation loss to zero:

$$\mathcal{L}_{\text{distill}} = 0. \quad (22)$$

Standard KD uses the unweighted mini-batch average:

$$\mathcal{L}_{\text{KD}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \ell_{\text{KD},i}. \quad (23)$$

Cross-Gated KD weights each per-window distillation term using watch signal quality together with teacher and watch-student confidence in the supervised class. Let

$$c_i^T = p_i^T(y_i), \quad c_i^w = p_i^w(y_i), \quad (24)$$

where $p_i^T = \text{softmax}(z_i^T)$ and $p_i^w = \text{softmax}(z_i^w)$ are the untempered class distributions. The raw gate weight is

$$\tilde{r}_i = q_i \min(c_i^T, c_i^w), \quad (25)$$

and the loss uses the detached weight $r_i = \text{sg}[\tilde{r}_i]$, so the gate acts as a sample weight. The Cross-Gated KD objective is

$$\mathcal{L}_{\text{gKD}} = \frac{\sum_{i \in \mathcal{B}} r_i \ell_{\text{KD},i}}{\sum_{i \in \mathcal{B}} r_i + \epsilon}, \quad (26)$$

where ϵ is a small constant for numerical stability. For Galaxy PPG, $q_i \in [0, 1]$ is the proportion of samples in the window whose device-reported watch-PPG status is valid. For WESAD, no corresponding wrist-BVP quality flag is available, so $q_i = 1$ for all windows and the gate depends only on teacher-student confidence. This conservative gate emphasizes distillation only when watch quality and both model confidences support the supervised class.

We additionally evaluate Teacher-gated and Student-gated variants as diagnostic ablations. Teacher-gated KD uses $r_i^T = \text{sg}[q_i c_i^T]$ whereas Student-gated KD uses $r_i^w = \text{sg}[q_i c_i^w]$. Each weight replaces r_i in Eq. 26. All teacher-guided and non-distilled variants produce the same Motion-aware watch-only inference path after training.

3.4 Joint Training and Watch-Only Inference

Training combines supervised stress classification, the selected privileged-supervision objective, and lightweight auxiliary losses. we set $w_i = 0.35 + 0.65q_i$,

$$\mathcal{L}_{\text{sup}} = \frac{\sum_{i \in \mathcal{B}} w_i (1 - p_{i,y_i})^{\gamma} \text{CE}_{\text{cls}}(z_i, y_i)}{\sum_{i \in \mathcal{B}} w_i + \epsilon}. \quad (27)$$

The watch classifier is trained from the ground-truth stress label using quality-weighted focal cross-entropy:

$$\mathcal{L}_{\text{watch}} = \mathcal{L}_{\text{sup}}(z^w, y). \quad (28)$$

When a privileged teacher is instantiated, its classifier is trained with the same supervised stress label:

$$\mathcal{L}_{\text{teacher}} = \mathcal{L}_{\text{sup}}(z^T, y). \quad (29)$$

Both supervised terms use focal cross-entropy with focusing parameter $\gamma = 1.5$ and label smoothing 0.05; class weights are computed within each training fold as $w_0 = 1$ and $w_1 = N_{\text{calm}}/N_{\text{stress}}$. The distillation term is selected from

$$\mathcal{L}_{\text{distill}} \in \{0, \mathcal{L}_{\text{KD}}, \mathcal{L}_{\text{gKD}}\}. \quad (30)$$

corresponding to non-distilled condition, Standard KD, and Cross-Gated KD. We use temperature $\tau = 4$.

Both datasets use a watch-side wavelet auxiliary objective. Let $\hat{\phi}^w$ denote the wavelet vector predicted from the deployable watch representation, and let ϕ^w denote the DWT band-energy-ratio target computed from the same watch PPG/BVP window. The auxiliary loss is

$$\mathcal{L}_{\text{wave}} = \text{SmoothL1}(\hat{\phi}^w, \phi^w). \quad (31)$$

Because the target is derived entirely from the deployable pulse signal, this objective does not introduce privileged sensing at inference.

Galaxy PPG additionally uses training-time auxiliary targets derived from non-watch sensors. When the Galaxy privileged branch is instantiated, the E4 BVP-ACC encoder has an auxiliary classifier trained with the stress label:

$$\mathcal{L}_{E4} = \mathcal{L}_{\text{sup}}(z^{E4}, y). \quad (32)$$

Galaxy PPG additionally uses Polar-derived rhythm and IBI targets as masked auxiliary supervision in teacher-guided privileged training. These targets supervise rhythm-prediction heads during training and are removed after training. The non-distilled Galaxy condition does not use E4 or Polar-derived rhythm auxiliary targets; it is trained only with the deployable watch task loss and the watch-side wavelet auxiliary loss. For teacher-guided Galaxy conditions, the rhythm objective averages the watch-side and teacher-side rhythm losses when both heads are present. WESAD does not use E4 or Polar-derived auxiliary targets.

For teacher-guided privileged-training configurations, the total objective is

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \lambda_w \mathcal{L}_{\text{watch}} + \lambda_T \mathcal{L}_{\text{teacher}} + \lambda_{\text{KD}} \mathcal{L}_{\text{distill}} + \lambda_{\text{wave}} \mathcal{L}_{\text{wave}} \\ & + \mathbb{I}_{\text{Galaxy}} (\lambda_{E4} \mathcal{L}_{E4} + \lambda_{\text{rhythm}} \mathcal{L}_{\text{rhythm}}), \end{aligned} \quad (33)$$

where $\mathbb{I}_{\text{Galaxy}}$ equals one for Galaxy PPG and zero for WESAD. We use

$$\lambda_w = 1.0, \quad \lambda_T = 0.80, \quad \lambda_{\text{KD}} = 0.08, \quad \lambda_{\text{wave}} = 0.05.$$

For Galaxy PPG, we set

$$\lambda_{E4} = 0.05, \quad \lambda_{\text{rhythm}} = 0.15.$$

For WESAD, the E4 and rhythm terms are absent. The non-distilled condition omits the privileged teacher branch and all teacher-dependent losses. Its objective is

$$\mathcal{L}_{\text{total}}^{\text{ND}} = \mathcal{L}_{\text{watch}} + \lambda_{\text{wave}} \mathcal{L}_{\text{wave}}, \quad (34)$$

The Galaxy-specific E4 and rhythm auxiliary terms are used only in teacher-guided privileged training.

In teacher-guided configurations, the teacher and watch branches are optimized jointly in each mini-batch. The teacher receives gradients from its supervised and applicable auxiliary losses. The teacher soft distribution used for distillation is detached, so Standard KD and Cross-Gated KD update only the deployable watch pathway through the distillation term.

After training, the privileged encoder, teacher classifier, E4 auxiliary classifier, teacher rhythm head, Polar-derived targets, privileged inputs, and all distillation computations are discarded. The deployed model evaluates only

$$h^w = f_w(x^{\text{pulse}}, x^{\text{acc}}, \phi^w, q), \quad z^w = C_w(h^w), \quad p^w = \text{softmax}(z^w). \quad (35)$$

Thus, all privileged-supervision objectives produce the same watch-only Motion-aware inference architecture. Additional implementation details are provided in Appendix A.

4 Experimental Setup

We evaluate watch-only stress recognition on two datasets. Galaxy PPG contains consumer smartwatch PPG and wrist ACC collected during a semi-naturalistic stress protocol, and is therefore close to the intended watch-only deployment setting. WESAD contains wrist BVP/ACC and richer chest-worn physiology collected under a controlled stress protocol, providing more windows and stronger privileged sensing. In both datasets, the deployed model receives only wrist pulse and wrist accelerometry, whereas richer physiological signals are available only during training. All methods within each dataset use the same label-independent preprocessing pipeline. Galaxy PPG is segmented into non-overlapping 20-s windows before window-level processing, whereas WESAD continuous streams are filtered, resampled, and normalized before overlapping 20-s windows are constructed. Galaxy watch PPG and WESAD wrist BVP are band-pass filtered between 0.5 and 8 Hz. Privileged ECG, respiration, and EMG are filtered between 0.5–40 Hz, 0.05–1 Hz, and 10–45 Hz, respectively, when those channels are available. Continuous-valued channels are robustly standardized using the channel-wise median and median absolute deviation, with no normalization statistics shared across participants. Wrist ACC inputs retain the three recorded axes and are augmented with acceleration magnitude. The deployable pathway also receives four watch-derived db4 level-4 DWT band-energy ratios computed only from the current PPG/BVP window. The watch-quality variable q_i is the valid-sample proportion from Galaxy device-reported PPG status flags; because WESAD has no equivalent wrist-BVP quality flag, $q_i = 1$ for all WESAD windows.

Using these two datasets and the same LOSO protocol, we organize the evaluation around two comparisons. The first comparison holds the downstream watch backbone fixed and varies how wrist ACC enters the deployable pulse-based model: Pulse-only sensing, Direct concatenation, learned Gated fusion, and the proposed Motion-aware encoder. The second comparison holds the deployed Motion-aware architecture fixed and varies only the training-time privileged-supervision objective: training without teacher-to-student distribution transfer, Standard KD, Cross-Gated KD, and the single-confidence gate variants. We report balanced accuracy, AUROC, F1, and the number of held-out subjects receiving single-class predictions under the validation-selected threshold. Additional fold-construction, bootstrap, and diagnostic-analysis details are provided in Appendix A.

4.1 Datasets and Prediction Tasks

As summarized in Table 1, Galaxy PPG represents the consumer watch setting. The deployable input consists of Samsung Galaxy Watch 5 PPG and wrist ACC, both processed at 25 Hz into 20-s non-overlapping windows. The binary task distinguishes baseline windows from Trier Social Stress Test preparation windows, yielding 23 participants and 393 unique physical windows. Training-time privileged information includes Empatica E4 BVP/ACC and Polar-derived rhythm/IBI targets. Polar ECG is not used as a raw teacher input; it is converted into masked auxiliary targets and removed at inference. WESAD provides a controlled multimodal benchmark. The deployable pathway uses wrist BVP and wrist ACC aligned to a 32-Hz watch representation and segmented into 20-s windows with a 10-s stride. The binary

Table 1. Summary of the two evaluation settings. Privileged signals are available during training only and are removed from the deployed model.

Dataset	Subj.	Calm windows	Stress windows	Window stride	Deployable signals	Privileged information
Galaxy PPG	23	195	198	20 s / 20 s	Watch PPG and wrist ACC	E4 BVP/ACC; Polar-derived rhythm/IBI targets
WESAD	15	1,742	977	20 s / 10 s	Wrist BVP and wrist ACC	Chest ACC, ECG, EMG, EDA, temperature, and respiration

task retains only baseline and stress conditions, producing 2,719 windows from 15 participants. The privileged teacher uses chest ACC, ECG, EMG, EDA, temperature, and respiration, all removed after training.

Together, the datasets provide two contrasting test conditions. Galaxy PPG is smaller but closer to the intended consumer-watch setting, whereas WESAD provides more windows and richer controlled physiology. We use both datasets to test whether ACC-integration and privileged-supervision effects are consistent across sensing configurations.

4.2 Compared Conditions

All compared conditions use the same dataset splits, preprocessing pipeline, checkpoint-selection criterion, validation-threshold protocol, and watch-only inference constraint within each dataset. The comparisons are organized to isolate two factors: how wrist ACC enters the deployable pulse-based model, and how training-time privileged supervision affects a fixed deployable Motion-aware architecture.

For the deployable ACC-integration comparison, we evaluate four backbone-matched watch variants. *Pulse-only* uses Galaxy watch PPG or WESAD wrist BVP and excludes the ACC pathway; we denote this condition as PPG-only on Galaxy PPG and BVP-only on WESAD. *Direct concat* encodes pulse and ACC separately and concatenates their learned representations. *Gated fusion* projects ACC features into the pulse feature space and learns a feature-level gate before passing the fused representation into the shared downstream backend. *Motion-aware* is the proposed deployable encoder, where ACC modulates and adjusts the pulse pathway. This comparison uses no privileged teacher-to-student distillation. We also train an ACC-only diagnostic model using wrist ACC axes and acceleration magnitude. It is used only to assess whether wrist motion alone carries protocol, activity, or sensing-condition information.

For the privileged-supervision comparison, we hold the deployed Motion-aware architecture, watch inputs, student-side supervised objectives and watch-only inference computation fixed. The compared conditions differ only in how training-time privileged predictions influence the student: training without teacher-to-student distribution transfer, Standard KD, Cross-Gated KD, and Teacher-gated or Student-gated variants. All privileged branches, auxiliary heads, and distillation computations are removed after training.

4.3 Evaluation Protocol and Metrics

All experiments use subject-disjoint leave-one-subject-out evaluation, with the held-out participant excluded from optimization, checkpoint selection, class-weight estimation, and threshold calibration. Within each dataset, all compared methods use the same splits. Model checkpoints are selected by validation AUROC. After checkpoint selection, a fold-specific decision threshold is chosen on the validation participants by maximizing balanced accuracy, and this threshold is then applied unchanged to the held-out test participant. We therefore report AUROC as a threshold-independent ranking metric, and balanced accuracy and positive-class F1 as threshold-dependent operating metrics. Metrics are

Table 2. ACC-integration comparison for deployable watch variants. Values are mean $\pm$ standard deviation across held-out participants; Single-class reports held-out subjects for which all test windows are assigned to one predicted class.

Dataset	Variant	BA	AUROC	F1	Single-class subjects
Galaxy PPG	PPG-only	0.613 ± 0.126	0.790 ± 0.183	0.614 ± 0.222	5/23 (21.7%)
	Direct concat	0.603 ± 0.189	0.757 ± 0.231	0.603 ± 0.232	4/23 (17.4%)
	Gated fusion	0.628 ± 0.193	0.719 ± 0.276	0.601 ± 0.275	5/23 (21.7%)
	Motion-aware	0.661 ± 0.179	0.727 ± 0.217	0.660 ± 0.221	4/23 (17.4%)
	ACC-only diagnostic	0.622 ± 0.099	0.672 ± 0.131	0.623 ± 0.117	0/23 (0.0%)
WESAD	BVP-only	0.801 ± 0.148	0.937 ± 0.081	0.723 ± 0.251	1/15 (6.7%)
	Direct concat	0.764 ± 0.161	0.910 ± 0.192	0.673 ± 0.263	1/15 (6.7%)
	Gated fusion	0.815 ± 0.164	0.919 ± 0.156	0.743 ± 0.257	1/15 (6.7%)
	Motion-aware	0.821 ± 0.149	0.939 ± 0.112	0.746 ± 0.259	1/15 (6.7%)
	ACC-only diagnostic	0.690 ± 0.178	0.784 ± 0.205	0.605 ± 0.230	1/15 (6.7%)

computed separately for each held-out participant and summarized as the mean and standard deviation across LOSO folds, treating the participant rather than the window as the evaluation unit. This subject-level reporting is important because recent stress-recognition work shows that strong within-dataset performance does not necessarily imply robust generalization across subjects or sensing scenarios [17].

We additionally report the number of held-out participants for which all test windows are assigned to a single predicted class. Because every evaluated participant contains both calm and stress windows, an all-calm or all-stress prediction represents a subject-level thresholding failure. For selected comparisons, we report paired subject-level differences with percentile-based (95%) confidence intervals computed by subject-clustered bootstrap resampling. All models are trained for at most 80 epochs using AdamW with an initial learning rate of 3×10^{-4} and weight decay 10^{-4} . Early stopping monitors validation AUROC with a patience of 10 epochs. Unless otherwise stated, experiments use random seed 42. Configurations without distillation, including Pulse-only, Direct concat, Gated fusion, and Motion-aware without teacher-to-student transfer, use batch size 32. Teacher-guided configurations use batch size 16 for Galaxy PPG and 24 for WESAD because the privileged teacher branch increases memory use. These batch sizes are fixed across the corresponding teacher-guided variants within each dataset.

5 Results

We present results for the two comparisons described above. The first comparison evaluates how wrist ACC enters a deployable pulse-based watch model while keeping the downstream backend fixed. The second comparison evaluates how privileged physiological supervision affects the same deployed Motion-aware architecture. Values are participant-level means and standard deviations across held-out participants. Balanced accuracy and F1 use validation-selected thresholds, whereas AUROC measures threshold-independent ranking.

5.1 ACC Integration in the Deployable Watch Model

We first compare the deployable ACC-integration variants. Across these variants, the downstream Refinement, temporal embedding, wavelet/quality fusion, and classifier are shared; the principal difference is whether ACC is excluded, concatenated as an unrestricted parallel representation, fused through a learned feature gate, or used to modulate and adjust the pulse pathway through the Motion-aware encoder.

Table 2 shows that ACC availability alone does not guarantee a better deployable pulse model. Under the matched downstream backbone, Direct concat yields lower balanced accuracy and F1 than the corresponding Pulse-only variant on both datasets. On Galaxy PPG, Direct concat obtains 0.603 balanced accuracy and 0.603 F1, compared with 0.613 and 0.614 for PPG only. On WESAD, Direct concat obtains 0.764 balanced accuracy and 0.673 F1, compared with 0.801 and 0.723 for BVP-only. Thus, using ACC as an unrestricted parallel input can hurt validation-thresholded performance relative to using pulse alone. The learned Gated-fusion baseline further clarifies this pattern. On Galaxy PPG, Gated fusion improves balanced accuracy over Direct concat, reaching 0.628, but its AUROC and F1 remain lower than the corresponding Direct-concat values in mean. On WESAD, Gated fusion is substantially stronger than Direct concat, reaching 0.815 balanced accuracy and 0.743 F1. This indicates that learned feature-level gating can repair part of the weakness of simple concatenation.

Among the ACC variants, Motion-aware achieves the highest mean balanced accuracy and F1 on both datasets. On Galaxy PPG, Motion-aware achieves 0.661 balanced accuracy and 0.660 F1, exceeding both Direct concat and Gated fusion. On WESAD, Gated fusion is competitive, but Motion-aware remains slightly higher in balanced accuracy and F1, reaching 0.821 and 0.746, respectively, and also preserves the strongest AUROC. These results show that Motion-aware conditioning improves over naive direct fusion and remains competitive even when the learned fusion baseline already performs strongly. AUROC and thresholded metrics capture different aspects of the watch model. On Galaxy PPG, the PPG-only model achieves the highest AUROC, indicating strong score ranking from pulse alone. However, Motion-aware achieves higher balanced accuracy and F1, suggesting that ACC pulse modeling yields a more usable validation-selected decision boundary under subject-disjoint evaluation. On WESAD, Motion-aware also achieves the highest AUROC while remaining slightly above Gated fusion in balanced accuracy and F1. Overall, the Motion-aware benefit is clearest in the thresholded deployment metrics, while AUROC shows that pulse-only ranking can remain strong even when its validation thresholded performance is weaker.

The ACC-only diagnostic shows that wrist motion alone carries nontrivial task signal, especially on Galaxy PPG. ACC-only achieves 0.622 balanced accuracy and 0.623 F1 on Galaxy, despite substantially lower AUROC than PPG-only. On WESAD, ACC-only is substantially weaker than all pulse-based variants, with 0.690 balanced accuracy and 0.605 F1. These diagnostics suggest that ACC may contain protocol, activity, or sensing-condition information, but should not be interpreted as direct physiological stress evidence. This supports using ACC as controlled context, not as an unrestricted substitute for pulse evidence.

Figure 3 provides a post-hoc view of where Motion-aware changes the probability assigned to the true class. On Galaxy PPG, the clearest gains over Direct concat occur in lower-validity windows, whereas gains relative to PPG-only are smaller and mixed, consistent with the strong AUROC of the pulse-only model. On WESAD, Motion-aware gains are positive across all BVP-only cells and across nearly all Direct-concat cells, indicating a broader improvement pattern in the controlled wrist-BVP benchmark. This post-hoc analysis is descriptive: it identifies where Motion-aware changes true-class probabilities, without attributing those changes to a specific internal mechanism.

5.2 Privileged Supervision on the Deployed Motion-Aware Model

Before comparing teacher-guided objectives, we first characterize the privileged teacher signal available during training. The Galaxy privileged teacher is informative but less stable across LOSO validation folds, with validation AUROC 0.832 ± 0.115 and fold-level range 0.637–1.000. The WESAD privileged teacher is stronger and more stable, with validation AUROC 0.990 ± 0.022 and minimum fold AUROC 0.934. We next evaluate whether privileged physiological supervision improves the deployed Motion-aware watch model. All compared objectives use the same watch-only

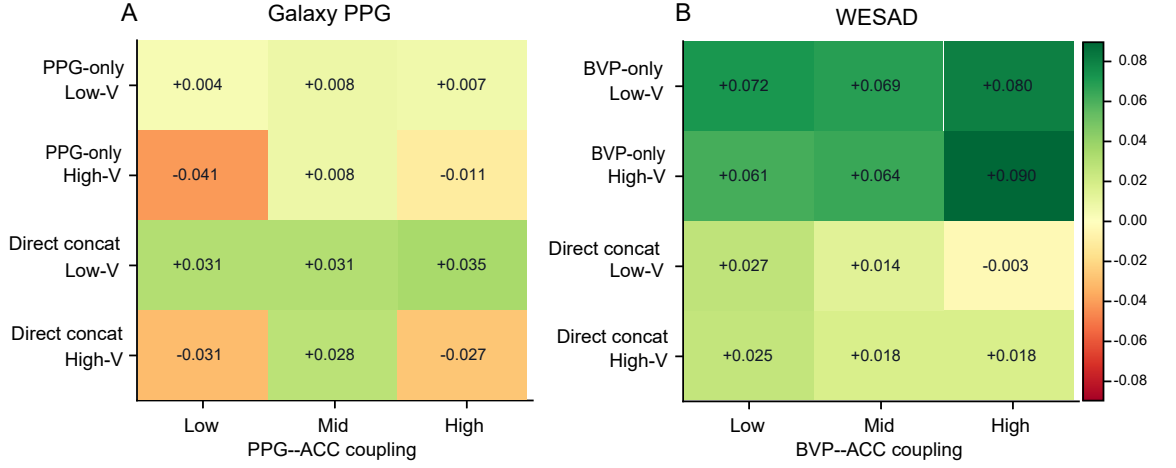


Fig. 3. Post-hoc true-class probability gains of Motion-aware encoding across dataset-relative sensing-condition groups. Each cell reports the mean change in probability assigned to the true class relative to the corresponding Pulse-only or Direct-concat baseline. Windows are grouped within each dataset by pulse-ACC coupling tertiles and by a post-hoc waveform-validity proxy. Low-V and High-V denote dataset-relative validity groups rather than absolute signal-quality thresholds.

Table 3. Privileged-teacher validation profile. Values summarize teacher-only validation AUROC across LOSO folds. These teacher models are diagnostic only: they are not deployable and are not used to select the distillation objective.

Dataset	Val. AUROC	Fold range / minimum
Galaxy PPG	0.832 ± 0.115	0.637–1.000
WESAD	0.990 ± 0.022	min. 0.934

Table 4. Privileged-supervision and confidence-source comparison on the deployed Motion-aware watch model. Values are participant-level mean $\pm$ standard deviation across LOSO folds. Balanced accuracy and F1 use the validation-selected threshold, whereas AUROC is threshold-independent. Teacher-gated and Student-gated KD isolate the confidence sources used by Cross-Gated KD.

Dataset	Objective	Balanced accuracy	AUROC	F1
Galaxy PPG	Non-distilled	0.661 ± 0.179	0.727 ± 0.217	0.660 ± 0.221
	Standard KD	0.611 ± 0.177	0.756 ± 0.201	0.574 ± 0.257
	Teacher gate	0.618 ± 0.170	0.749 ± 0.207	0.618 ± 0.229
	Student gate	0.640 ± 0.169	0.744 ± 0.206	0.632 ± 0.228
	Cross-Gated KD	0.640 ± 0.178	0.762 ± 0.190	0.629 ± 0.244
WESAD	Non-distilled	0.821 ± 0.149	0.939 ± 0.112	0.746 ± 0.259
	Standard KD	0.820 ± 0.139	0.944 ± 0.098	0.743 ± 0.253
	Teacher gate	0.788 ± 0.147	0.946 ± 0.091	0.706 ± 0.258
	Student gate	0.790 ± 0.154	0.947 ± 0.088	0.697 ± 0.281
	Cross-Gated KD	0.790 ± 0.140	0.942 ± 0.089	0.694 ± 0.263

inference architecture; they differ only in how teacher-to-student distribution transfer is used during training. Table 4 compares non-distilled training, Standard KD, Cross-Gated KD, and the single-confidence gate variants on the same

deployed Motion-aware watch model. AUROC measures threshold-independent ranking, whereas balanced accuracy and F1 measure performance after applying the validation-selected threshold.

On Galaxy PPG, the non-distilled Motion-aware model achieves the highest balanced accuracy and F1, with values of 0.661 and 0.660. Standard KD increases AUROC from 0.727 to 0.756, but reduces balanced accuracy to 0.611 and F1 to 0.574. Cross-Gated KD partially mitigates this thresholded-performance loss, reaching 0.640 balanced accuracy and 0.629 F1, but it still does not exceed the non-distilled model on balanced accuracy or F1. Thus, on Galaxy PPG, privileged supervision improves score ranking but can weaken the validation-thresholded decision behavior. On WESAD, Standard KD behaves differently. The non-distilled model achieves 0.821 balanced accuracy, 0.939 AUROC, and 0.746 F1, while Standard KD yields 0.820 balanced accuracy, 0.944 AUROC, and 0.743 F1. This indicates that, when the privileged teacher is strong and stable, uniform distillation can preserve thresholded performance while slightly improving ranking. Cross-Gated KD does not improve this trade-off: it keeps AUROC close to the other teacher-guided condition but lowers balanced accuracy and F1 to 0.790 and 0.694. In this stronger-teacher regime, conservative confidence weighting appears to suppress useful teacher information, weakening transfer.

The single-confidence variants clarify this behavior. On Galaxy PPG, Teacher-gated KD does not recover the threshold-dependent degradation introduced by Standard KD. Student-gated KD yields higher balanced accuracy and F1 than Teacher-gated KD, suggesting that support from the deployable watch pathway is important when deciding whether privileged guidance is compatible with the student representation. Cross-Gated KD is similar to Student-gated KD in balanced accuracy and F1, while producing the highest AUROC among the gated variants. WESAD shows the opposite pattern. Teacher-gated, Student-gated, and Cross-Gated KD all reduce balanced accuracy and F1 relative to the non-distilled condition and Standard KD, despite the strong WESAD privileged teacher. Thus, confidence weighting is not a universal reliability mechanism. Teacher confidence reflects support in the privileged sensing space, and student confidence reflects support in the deployable watch representation, but neither directly guarantees that teacher evidence can be transferred without disturbing the validation-selected operating point.

Privileged supervision does not produce a uniform improvement across datasets or metrics. On Galaxy PPG, Standard KD improves AUROC but harms balanced accuracy and F1, while Cross-Gated KD recovers part of the threshold-dependent loss. On WESAD, Standard KD closely matches the non-distilled condition on balanced accuracy and F1 and slightly improves AUROC, whereas Cross-Gated KD is overly conservative. These results show that teacher quality alone does not determine the value of a distillation objective. The effect also depends on whether privileged evidence can be transferred into the watch representation without disturbing its validation-thresholded operating behavior. The non-distilled condition therefore remains a competitive first-class choice, and the value of Standard or confidence-weighted KD must be evaluated against both ranking and threshold-dependent deployment metrics.

5.3 Subject-Level Uncertainty and Deployment Diagnostics

Table 5 reports paired subject-level differences for the principal ACC-integration and privileged-supervision comparisons. These subject-clustered bootstrap intervals complement the aggregate LOSO means by summarizing how consistently the mean differences appear across held-out participants.

For the ACC-integration comparison, the paired intervals support the main aggregate pattern. Gated fusion improves over Direct concat most clearly on WESAD, where the balanced-accuracy and F1 intervals exclude zero. Motion-aware has positive mean differences relative to Gated fusion on both datasets, although the corresponding intervals overlap zero. This pattern supports Motion-aware conditioning as a strong aggregate design while showing that its advantage over learned gated fusion is not uniform across held-out participants. For the privileged-supervision comparison, the

Table 5. Paired subject-level differences for the principal ACC-integration and privileged-supervision comparisons. Each entry reports the mean difference between methods (A-B), followed by its 95% percentile-bootstrap confidence interval based on 20,000 resamples of held-out subjects. Bold entries indicate intervals that exclude zero.

Dataset	Paired comparison (A-B)	Δ BA [95% CI]	Δ AUROC [95% CI]	Δ F1 [95% CI]
Galaxy	Motion-aware – PPG-only	+0.048 [−0.008, +0.107]	−0.062 [−0.120, −0.010]	+0.046 [−0.009, +0.100]
Galaxy	Motion-aware – Direct concat	+0.058 [−0.004, +0.127]	−0.030 [−0.100, +0.047]	+0.057 [−0.001, +0.118]
WESAD	Motion-aware – BVP-only	+0.020 [−0.029, +0.076]	+0.002 [−0.040, +0.039]	+0.023 [−0.038, +0.089]
WESAD	Motion-aware – Direct concat	+0.057 [+0.004, +0.113]	+0.030 [−0.003, +0.080]	+0.073 [−0.003, +0.151]
Galaxy	Gated fusion – Direct concat	+0.025 [−0.074, +0.117]	−0.039 [−0.174, +0.074]	−0.002 [−0.106, +0.091]
Galaxy	Motion-aware – Gated fusion	+0.033 [−0.063, +0.137]	+0.009 [−0.114, +0.154]	+0.059 [−0.039, +0.173]
WESAD	Gated fusion – Direct concat	+0.051 [+0.019, +0.088]	+0.009 [−0.011, +0.035]	+0.069 [+0.026, +0.118]
WESAD	Motion-aware – Gated fusion	+0.006 [−0.027, +0.044]	+0.020 [−0.003, +0.051]	+0.003 [−0.048, +0.060]
Galaxy	Standard KD – Non-distilled	−0.050 [−0.107, −0.003]	+0.027 [−0.019, +0.071]	−0.086 [−0.154, −0.025]
Galaxy	Cross-Gated KD – Standard KD	+0.030 [−0.010, +0.079]	+0.008 [−0.013, +0.032]	+0.055 [+0.004, +0.117]
Galaxy	Cross-Gated KD – Non-distilled	−0.021 [−0.057, +0.013]	+0.035 [−0.008, +0.080]	−0.031 [−0.068, +0.002]
WESAD	Standard KD – Non-distilled	−0.002 [−0.030, +0.024]	+0.004 [−0.010, +0.022]	−0.003 [−0.034, +0.026]
WESAD	Cross-Gated KD – Standard KD	−0.030 [−0.061, +0.001]	−0.002 [−0.016, +0.012]	−0.049 [−0.092, −0.007]
WESAD	Cross-Gated KD – Non-distilled	−0.032 [−0.062, −0.006]	+0.003 [−0.010, +0.019]	−0.052 [−0.095, −0.015]

Table 6. Inference-time footprint of deployable watch variants. Parameter counts include only the deployed watch pathway. CPU latency is measured for one 20-s window with batch size one after warm-up, excluding preprocessing, privileged teacher branches, auxiliary heads, and distillation computations. Galaxy PPG and WESAD use watch input lengths of 500 and 640 samples, respectively.

Variant	Parameters	Galaxy PPG latency	WESAD latency
Pulse-only	1.54M	10.925 ± 1.605 ms	12.390 ± 2.186 ms
Direct concat	1.70M	14.616 ± 0.817 ms	17.182 ± 2.440 ms
Gated fusion	1.80M	15.629 ± 1.910 ms	16.311 ± 1.215 ms
Motion-aware	2.23M	16.969 ± 1.241 ms	18.220 ± 1.325 ms

paired intervals highlight the conditional value of teacher guidance. On Galaxy PPG, Standard KD improves AUROC in mean but reduces balanced accuracy and F1 relative to the non-distilled condition, with the balanced-accuracy and F1 intervals excluding zero. Cross-Gated KD improves F1 relative to Standard KD but remains below the non-distilled condition on thresholded metrics. On WESAD, Standard KD remains close to the non-distilled condition, whereas Cross-Gated KD reduces F1 relative to Standard KD and reduces both balanced accuracy and F1 relative to the non-distilled condition. These patterns show that privileged teacher guidance can improve score ranking while changing validation-thresholded decisions in a less favorable direction.

The paired intervals summarize behavior at the validation-selected threshold. To examine nearby threshold choices, we perturb each fold-specific threshold by fixed offsets $\Delta\delta \in \{-0.10, -0.05, 0, 0.05, 0.10\}$, clipping the resulting value to $[0, 1]$. Figure 4 shows that the ordering of privileged-supervision objectives changes within this threshold neighborhood. At $\Delta\delta = 0$, the curves match the main validation-thresholded results: the non-distilled condition is strongest on Galaxy PPG for balanced accuracy and F1, while the non-distilled condition and Standard KD are closely matched on WESAD. With larger threshold offsets, several F1 orderings change. This analysis reinforces the distinction between ranking quality and deployment-time thresholded behavior: AUROC gains should be interpreted together with balanced accuracy and F1 after threshold selection.

Table 6 reports the inference-time footprint of the deployable ACC-integration variants. Motion-aware has the largest deployed watch pathway, increasing the parameter count from 1.54M for Pulse-only to 2.23M. CPU latency remains below 19 ms per 20-s window in both evaluation settings. The higher WESAD latency is consistent with its

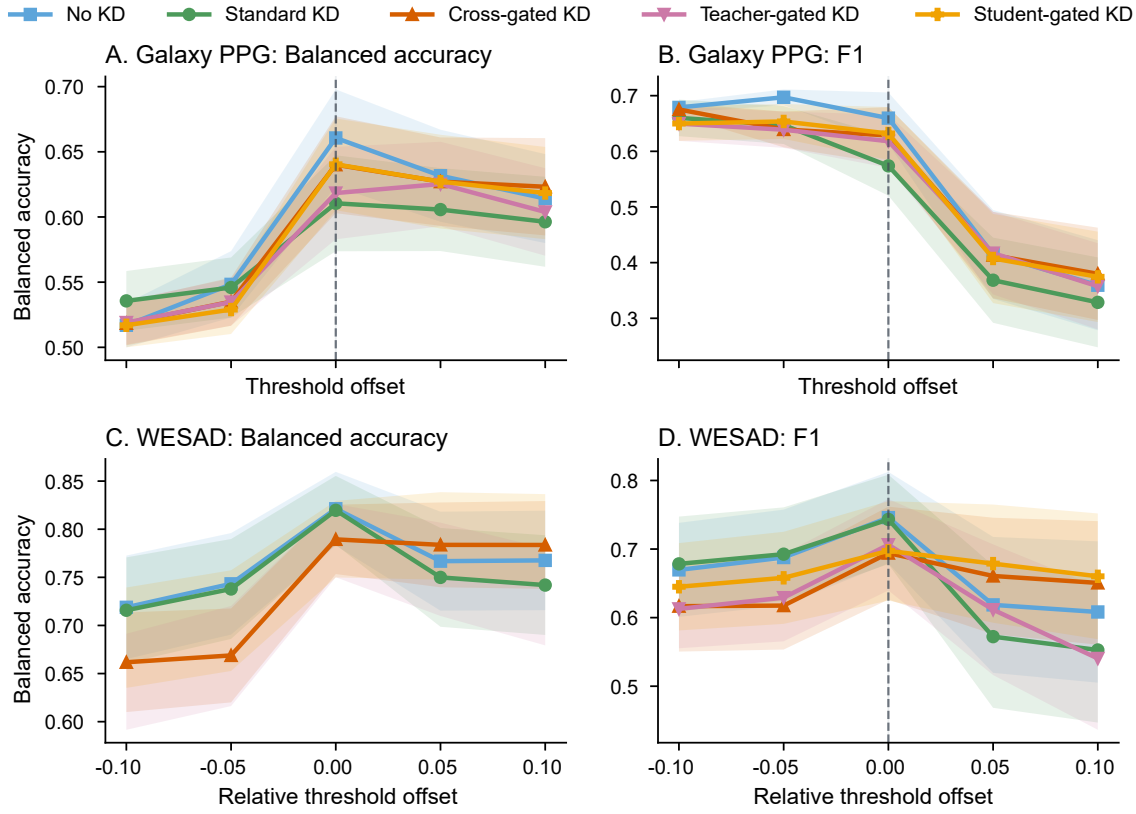


Fig. 4. Threshold robustness of privileged-supervision objectives on Galaxy PPG and WESAD. For each held-out participant and objective, the validation-selected threshold δ^* is perturbed by fixed offsets $\Delta\delta \in \{-0.10, -0.05, 0, 0.05, 0.10\}$. The vertical dashed line denotes the original operating point ($\Delta\delta = 0$). Markers show the mean across held-out participants, and shaded regions denote mean $\pm$ SEM.

longer watch input length. These measurements characterize computational footprint rather than optimized on-watch deployment. All privileged-supervision objectives differ only during training; after teacher branches, auxiliary heads, and distillation computations are removed, they share the same Motion-aware watch-only inference path and have identical inference-time footprints.

6 Discussion

The results highlight a deployment-aware principle for watch-only stress recognition: retained and training-only signals should be used according to their inference-time roles. PPG/BVP provides the primary pulse-related physiological evidence, whereas wrist ACC is most useful as context for interpreting that evidence rather than as an unrestricted parallel prediction signal. Likewise, richer physiology can guide training, but its value depends on whether privileged teacher evidence transfers to the deployed watch representation. Together, these findings show that both motion and privileged physiology are useful only when their functional role matches the watch-only inference constraint.

6.1 Motion as Context for Pulse Interpretation

The ACC-integration comparison shows that wrist motion is informative but not automatically beneficial. Direct pulse+ACC concatenation reduces validation-thresholded balanced accuracy and F1 relative to Pulse-only sensing on both Galaxy PPG and WESAD. The ACC-only diagnostic further shows that wrist motion alone carries nontrivial task signal, especially on Galaxy PPG. Together, these results suggest that ACC can encode activity, posture, protocol, or sensing-condition information, but using it as an unrestricted predictive pathway can weaken the final watch decision.

The Motion-aware encoder addresses this by assigning ACC a contextual role. Pulse remains the primary physiological pathway, while ACC affects the pulse representation through bounded Adaptation and Correction. This design keeps the deployed representation grounded in pulse evidence rather than treating ACC as an interchangeable predictive modality. Across the evaluated deployable variants, Motion-aware encoding achieves the strongest mean balanced accuracy and F1. The gain is clearest against Direct concat, where Motion-aware avoids the degradation caused by unrestricted fusion. Gated fusion remains a competitive learned fusion baseline, especially on WESAD, but Motion-aware remains slightly higher in mean balanced accuracy and F1 when the remaining watch-model components are kept fixed.

The post-hoc behavior analysis is consistent with this interpretation. On Galaxy PPG, Motion-aware gains over Direct concat are most evident in lower-validity windows, where pulse interpretation is likely more affected by sensing conditions. On WESAD, gains are broader across sensing-condition groups. These analyses describe where the model changes true-class probabilities; they do not establish explicit artifact removal. Because the correction mask and residual basis are not supervised by artifact annotations or clean-waveform targets, Adaptation and Correction should be interpreted as learned feature-level pulse adjustment.

6.2 Ranking, Thresholding, and Privileged Guidance

The results also show that AUROC and validation-thresholded metrics capture different aspects of watch-only deployment. AUROC measures score ranking across thresholds, whereas balanced accuracy and F1 measure the behavior after a threshold is selected on validation participants and transferred to a held-out participant. A model can improve ranking while still producing a less transferable operating point.

This distinction appears in both experimental comparisons. On Galaxy PPG, PPG-only achieves the highest AUROC, but Motion-aware yields stronger balanced accuracy and F1. In the privileged-supervision comparison, Standard KD improves Galaxy AUROC while reducing validation-thresholded balanced accuracy and F1 relative to the non-distilled Motion-aware model. Cross-Gated KD mitigates part of this loss, but does not surpass the non-distilled condition on thresholded metrics. On WESAD, where the privileged teacher is stronger and more stable, Standard KD largely preserves thresholded performance while slightly improving AUROC. Confidence-gated objectives are less favorable in this setting, suggesting that conservative weighting can suppress useful teacher guidance when the teacher signal is reliable.

These results argue against treating privileged physiology as an oracle. Teacher confidence reflects support in the privileged sensing space, whereas student confidence reflects support in the deployable watch representation. This same confidence weighting can reduce harmful transfer on Galaxy PPG, where teacher guidance is less stable, but become overly conservative on WESAD, where the teacher signal is stronger. The practical implication is that the non-distilled Motion-aware model should remain a primary deployment baseline, and privileged supervision should be evaluated using the metrics relevant to the intended operating point, not only by teacher strength or AUROC gains. A

reliable criterion for choosing between non-distilled training, uniform distillation, and confidence-weighted distillation remains an open modeling problem.

6.3 Deployment Scope and Limitations

All privileged-supervision objectives modify training only. At inference, the deployed model uses watch-available pulse, wrist ACC, watch-derived wavelet features, and the watch-quality variable. Privileged encoders, teacher classifiers, rhythm targets, auxiliary heads, and distillation computations are removed after training. The footprint analysis indicates that the Motion-aware pathway remains modest in this implementation, with CPU latency below the scale of a 20-s input window. These measurements characterize computational footprint; optimized smartwatch deployment still requires hardware-specific implementation and evaluation [5].

The study has several limitations. Galaxy PPG is closest to the intended consumer-watch setting, but it has a small cohort and protocol-derived labels, so session structure, activity, or other contextual cues may contribute to performance. WESAD provides more windows and richer controlled physiology, but it does not capture free-living behavioral variability. The two datasets differ in device type, protocol, sample size, label construction, and privileged sensing; the cross-dataset comparison therefore supports robustness across contrasting evaluation settings, but does not isolate individual dataset factors. Finally, the evaluated watch architectures are complete practical designs rather than parameter-matched causal ablations, so the observed gains reflect the combined effect of representation constraints, model capacity, and training objectives. Future work should evaluate broader populations, repeated training runs, additional watch platforms, and free-living datasets with independent stress labels.

7 Conclusion

This work studied watch-only stress recognition under asymmetric sensing: richer physiological signals may be available during training, while inference is restricted to watch pulse and wrist acceleration. Across Galaxy PPG and WESAD, wrist ACC is most useful when it supports pulse interpretation rather than entering as an unrestricted parallel prediction signal. The Motion-aware encoder uses ACC to adapt and adjust the pulse pathway and achieves the strongest mean balanced accuracy and F1 among the evaluated deployable variants.

Privileged physiological supervision provides conditional training guidance rather than uniformly beneficial transfer. Standard KD can improve AUROC while weakening validation-thresholded behavior, and confidence weighting helps only when its conservatism matches the teacher–student relationship. All evaluated privileged-supervision objectives leave the same watch-only inference path after training. Overall, the results support a deployment-aware approach: motion should be modeled as context for pulse evidence, and privileged physiology should be evaluated as conditional guidance under the constraints of watch-only inference.

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A Implementation Details

A.1 Signal Processing and Watch-Derived Features

All compared methods within each dataset use the same preprocessed windows and watch-derived features. Galaxy PPG is processed at 25 Hz using non-overlapping 20-s windows. WESAD wrist BVP and ACC are aligned to a 32-Hz watch representation and segmented into 20-s windows with a 10-s stride. Only baseline and stress windows are retained for both datasets.

Galaxy watch PPG and WESAD wrist BVP are band-pass filtered between 0.5 and 8 Hz. When privileged channels are used during training, ECG is filtered between 0.5 and 40 Hz, respiration between 0.05 and 1 Hz, and EMG between 10 and 45 Hz. EDA and temperature are resampled to the teacher representation without the pulse-specific band-pass filter. Continuous-valued channels are standardized as

$$\tilde{x} = \frac{x - \text{median}(x)}{1.4826 \text{ MAD}(x) + \epsilon}, \quad (36)$$

where ϵ is used for numerical stability. Galaxy statistics are computed within each extracted window. WESAD statistics are computed separately for each participant and channel before window construction.

For wrist accelerometry, the deployable input keeps the three recorded axes and adds the magnitude channel

$$a_{\text{mag}}(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2}. \quad (37)$$

Thus, the ACC input contains $(a_x, a_y, a_z, a_{\text{mag}})$. The watch-derived wavelet vector is computed only from the deployable PPG/BVP window. We apply a level-4 discrete wavelet transform with the Daubechies-4 wavelet and use normalized band-energy ratios from A_4 , D_4 , D_2 , and D_1 . For band b with coefficients $c_{b,k}$,

$$e_b = \frac{\sum_k c_{b,k}^2}{\sum_{b'} \sum_k c_{b',k}^2 + \epsilon}. \quad (38)$$

The wavelet feature vector is

$$\phi^w = [e_{A_4}, e_{D_4}, e_{D_2}, e_{D_1}]. \quad (39)$$

For Galaxy PPG, Polar H10 ECG is converted into auxiliary rhythm targets rather than used as a raw teacher input. The target vector for window i is

$$\mathbf{r}_i = [\text{HR}_i, \text{IBI}_i, \text{VarIBI}_i], \quad (40)$$

with an element-wise validity mask $\mathbf{m}_i$. These targets are used only for masked auxiliary supervision during training and are removed at inference.

A.2 Training Objectives and Auxiliary Losses

All supervised stress-classification terms use a quality-weighted focal cross-entropy. For a mini-batch B , the per-window weight is

$$w_i = 0.35 + 0.65 \text{clip}(q_i, 0, 1),$$

and

$$\mathcal{L}_{\text{sup}} = \frac{\sum_{i \in B} w_i (1 - p_{i,y_i})^\gamma \text{CE}_{\text{cls}}(z_i, y_i)}{\sum_{i \in B} w_i + \epsilon}.$$

We use $\gamma = 1.5$, label smoothing 0.05, and fold-specific class weights. All supervised stress-classification losses use focal cross-entropy with focusing parameter $\gamma = 1.5$, label smoothing 0.05, and fold-specific class weights. For each training

fold, class weights are computed from the training participants only:

$$w_0 = 1, \quad w_1 = \frac{N_{\text{calm}}}{N_{\text{stress}}}, \quad (41)$$

where N_{calm} and N_{stress} are the numbers of calm and stress training windows.

The watch-side wavelet auxiliary loss predicts the deployable DWT feature vector ϕ^w from the watch representation. For mini-batch $\mathcal{B}$, we use

$$\mathcal{L}_{\text{wave}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \text{SmoothL1}(\hat{\phi}_i^w, \phi_i^w). \quad (42)$$

Because ϕ_i^w is computed from the deployable PPG/BVP window, this auxiliary objective does not use privileged sensing.

For Galaxy PPG, Polar-derived rhythm targets are used as masked auxiliary supervision. Let

$$\mathbf{r}_i = [\text{HR}_i, \text{IBI}_i, \text{VarIBI}_i] \quad (43)$$

denote the rhythm target vector for window i , and let $\mathbf{m}_i \in \{0, 1\}^3$ indicate which target entries are valid. The watch-side rhythm loss is

$$\mathcal{L}_{\text{rhythm}}^w = \frac{\sum_{i \in \mathcal{B}} \sum_{d=1}^3 m_{i,d} \text{SmoothL1}(\hat{r}_{i,d}^w, r_{i,d})}{\sum_{i \in \mathcal{B}} \sum_{d=1}^3 m_{i,d} + \epsilon}. \quad (44)$$

When a Galaxy privileged teacher is present, the teacher-side rhythm loss is defined analogously:

$$\mathcal{L}_{\text{rhythm}}^T = \frac{\sum_{i \in \mathcal{B}} \sum_{d=1}^3 m_{i,d} \text{SmoothL1}(\hat{r}_{i,d}^T, r_{i,d})}{\sum_{i \in \mathcal{B}} \sum_{d=1}^3 m_{i,d} + \epsilon}. \quad (45)$$

For the non-distilled Galaxy condition, the rhythm term is $\mathcal{L}_{\text{rhythm}}^w$. For teacher-guided Galaxy conditions, we use

$$\mathcal{L}_{\text{rhythm}} = \frac{1}{2} (\mathcal{L}_{\text{rhythm}}^w + \mathcal{L}_{\text{rhythm}}^T). \quad (46)$$

WESAD does not use Polar-derived rhythm or IBI auxiliary targets.

For Galaxy PPG, the privileged E4 branch also has an auxiliary stress classifier:

$$\mathcal{L}_{E4} = \text{FocalCE}(\mathbf{z}^{E4}, y). \quad (47)$$

This auxiliary classifier is used only during training and is removed from the deployed watch-only model.

The loss weights used in all experiments are

$$\lambda_w = 1.0, \quad \lambda_T = 0.80, \quad \lambda_{\text{KD}} = 0.08, \quad \lambda_{\text{wave}} = 0.05. \quad (48)$$

For Galaxy PPG, the additional auxiliary weights are

$$\lambda_{E4} = 0.05, \quad \lambda_{\text{rhythm}} = 0.15. \quad (49)$$

For WESAD, the E4 and rhythm terms are absent.

A.3 LOSO Splits, Thresholding, and Bootstrap

All experiments use subject-disjoint leave-one-subject-out evaluation. For each outer fold, one participant is held out as the test participant. A validation participant is selected from the remaining participants using a fixed cyclic rule, and all other participants are used for training. The held-out test participant is not used for optimization, checkpoint selection, class-weight estimation, or decision-threshold calibration.

Model checkpoints are selected by validation AUROC. After checkpoint selection, a fold-specific decision threshold is chosen on the validation participant by maximizing balanced accuracy:

$$\delta^* = \arg \max_{\delta} \text{BA}_{\text{val}}(\delta). \quad (50)$$

The selected threshold is then applied unchanged to the held-out test participant:

$$\hat{y}_i = \mathbb{I} [p_i^w(y = 1) \geq \delta^*]. \quad (51)$$

For each held-out participant, we compute AUROC, balanced accuracy, and positive-class F1. Balanced accuracy is

$$\text{BA} = \frac{1}{2} (\text{TPR} + \text{TNR}). \quad (52)$$

Participant-level metric values are then summarized as mean and standard deviation across LOSO folds.

We also report single-class prediction failures. For held-out participant s with test set $\mathcal{T}_s$, a method is counted as single-class if

$$\sum_{i \in \mathcal{T}_s} \hat{y}_i = 0 \quad \text{or} \quad \sum_{i \in \mathcal{T}_s} \hat{y}_i = |\mathcal{T}_s|. \quad (53)$$

Because every evaluated participant contains both calm and stress windows, this indicates a subject-level thresholding failure.

For selected paired comparisons, we compute subject-level paired differences. Let m_s^A and m_s^B denote the metric values for methods A and B on held-out participant s . The paired difference is

$$d_s = m_s^A - m_s^B. \quad (54)$$

We report the mean paired difference and a percentile-based 95% confidence interval using subject-clustered bootstrap resampling. Each bootstrap replicate samples held-out participants with replacement, computes the mean of d_s over the resampled participants, and the 2.5th and 97.5th percentiles of the bootstrap distribution are used as the confidence interval.